

CSE 5825: Probabilistic Graph Analysis

Homework 1

Problem 1 (10 points)

Prove

$$p(x, y | z) = p(x | z) p(y | x, z)$$

and also

$$p(x | y, z) = \frac{p(y | x, z) p(x | z)}{p(y | z)}.$$

$$p(x | y, z) = \frac{p(y | x, z) p(x | z)}{p(y | x, z) p(x | z) + p(y | x, z) p(x | z)}.$$

Problem 2 (15 points)

A college has three departments, A, B, and C, with female percentages 35%, 20%, and 55% respectively. A department is chosen at random, then a student is chosen at random from that department.

- (a) $P(\text{female})$
- (b) $P(\text{Dept} = B | \text{female})$
- (c) $P(\text{Dept} = A | \text{male})$

Problem 3 (20 points)

Consider $P(A, B, C, D)$ where A, B are binary and C, D have 3 values each.

- (a) Parameters for the full joint
- (b) Parameters assuming $P(A, B, C, D) = P(A)P(B | A)P(C | B)P(D | B)$
- (c) Network diagram and independence questions

Is $A \perp B$? Is $A \perp B | C$?

Problem 4 (30 points)

Chest Clinic belief network. Variables: x = positive X-ray, d = dyspnea, e = either tuberculosis or lung cancer, t = tuberculosis, l = lung cancer, b = bronchitis, a = visited Asia, s = smoker.

(a) Conditional independence statements (true/false)

1. tuberculosis $\perp$ smoking | shortness of breath
2. lung cancer $\perp$ bronchitis | smoking
3. visit to Asia $\perp$ smoking | lung cancer
4. visit to Asia $\perp$ smoking | lung cancer, shortness of breath

(b) $p(d)$, $p(d \mid s = \text{tr})$, $p(d \mid s = \text{fa})$

Computed via code (see `chest_clinic_skeleton.ipynb`, submitted alongside this file). Factorization used:

$$p(a, t, s, l, b, e, x, d) = p(a) p(t \mid a) p(s) p(l \mid s) p(b \mid s) p(e \mid t, l) p(x \mid e) p(d \mid e, b).$$

Skeleton of the computation (fill in CPT values and complete the marked steps in the notebook, then transcribe the final numbers here):

Algorithm 1 Marginal and conditional queries on the Chest Clinic network

```
1: procedure JOINTPROB( $a, t, s, l, b, e, x, d$ )
2:   return  $p(a) \cdot p(t \mid a) \cdot p(s) \cdot p(l \mid s) \cdot p(b \mid s) \cdot p(e \mid t, l) \cdot p(x \mid e) \cdot p(d \mid e, b)$ 
3: end procedure
4: procedure MARGINALD
5:    $p_d[\text{tr}] \leftarrow 0$ ,    $p_d[\text{fa}] \leftarrow 0$ 
6:   for each assignment of  $a, t, s, l, b, e, x, d$  do
7:      $p_d[d] \leftarrow p_d[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
8:   end for
9:   return  $p_d$ 
10: end procedure
11: procedure CONDITIONALDGIVENS( $s^*$ )
12:    $u[\text{tr}] \leftarrow 0$ ,    $u[\text{fa}] \leftarrow 0$ 
13:   for each assignment of  $a, t, s, l, b, e, x, d$  with  $s = s^*$  do
14:      $u[d] \leftarrow u[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
15:   end for
16:   return  $u$  normalized so  $u[\text{tr}] + u[\text{fa}] = 1$ 
17: end procedure
```

$$\begin{aligned} p(d) &= \\ p(d \mid s = \text{tr}) &= \\ p(d \mid s = \text{fa}) &= \end{aligned}$$

Problem 5 (15 points)

Consider $p(x, y, w, z) = p(z \mid w) p(w \mid x, y) p(x) p(y)$.

(a) Network diagram

(b) $p(x \mid z)$

(c) $p(y \mid z)$

(d) Condition for $x \perp y \mid z$

Problem 6 (15 points)

Graph: $A \rightarrow C, A \rightarrow E, B \rightarrow C, B \rightarrow D, C \rightarrow E$.

(a) Independent pairs, no observations

(b) Independent pairs given E

(c) Decomposition of $P(A, B, C, E)$

(d) Parameters for CPD of E (A, B, D binary; C, E three values)